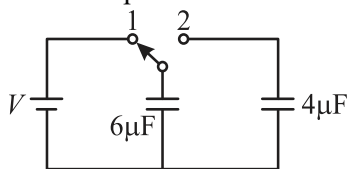


SECTION-I (PHYSICS)

Single Correct Type Questions

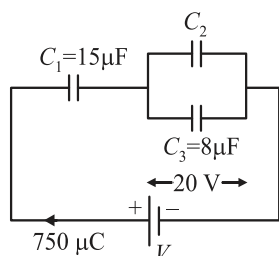
1. A $6\ \mu\text{F}$ capacitor is first charged by connecting it to a battery (position 1). When the switch is moved to position 2, the capacitor is connected to an initially uncharged capacitor of capacitance $4\ \mu\text{F}$. What percentage of the stored energy in the $6\ \mu\text{F}$ capacitor is dissipated when the switch is moved to position 2?



- (1) 20% (2) 40%
(3) 50% (4) 60%
2. Four charges Q are placed at the vertices of a regular tetrahedron. Find the force on charge placed at one of the vertices of this tetrahedron of side length d .

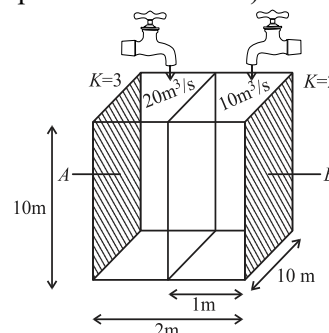
(1) $\sqrt{7} \frac{Q^2}{4\pi\epsilon_0 d^2}$ (2) $\sqrt{6} \frac{Q^2}{4\pi\epsilon_0 d^2}$
(3) $\sqrt{3} \frac{Q^2}{4\pi\epsilon_0 d^2}$ (4) $\sqrt{2} \frac{Q^2}{4\pi\epsilon_0 d^2}$

3. Find the value of capacitance C_2 in the shown circuit.

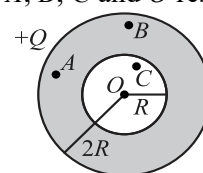


- (1) $\frac{59}{2}\ \mu\text{F}$ (2) $\frac{45}{2}\ \mu\text{F}$
(3) $\frac{65}{3}\ \mu\text{F}$ (4) $16\ \mu\text{C}$
4. Two metallic plates are arranged with a thin conducting sheet placed vertically between them, forming a symmetric structure. The sides not covered by the metals are enclosed using perfect insulators, turning the assembly into a chamber-like setup. Two inlets connected to reservoirs are opened simultaneously at time $t = 0$, allowing different non-conductive liquids to fill the chamber. After 5 seconds, both inlets are sealed. Given the rate at which the liquids enter (in volume per second) and their respective relative permittivities (K) as in the shown diagram, determine the effective capacitance measured between terminals A and B after the system is

sealed. (ϵ_0 is permittivity of free space) (Assume plates separation to be small)



- (1) $75\epsilon_0\ \text{F}$ (2) $100\epsilon_0\ \text{F}$
(3) $55\epsilon_0\ \text{F}$ (4) $60\epsilon_0\ \text{F}$
5. A body having mass m and carrying a positive charge $+q$ is launched with speed v in a region devoid of gravitational influence. An electric field, E is present in the region, oriented along the negative direction of the y -axis. If the body is initially projected in x - y plane at an angle θ with respect to the x -axis, determine the initial rate at which its speed changes.
- (1) $\frac{qE}{m} \operatorname{cosec} \theta$ (2) $-\frac{qE}{m} \sec \theta$
(3) $-\frac{qE}{m} \sin \theta$ (4) $\frac{qE}{m} \cos \theta$
6. A capacitor with capacitance C is initially charged to voltage V_0 and is isolated from any circuit. Subsequently, this charged capacitor is connected in parallel with an uncharged capacitor having capacitance $\frac{C}{2}$. Determine the energy dissipated when charge equilibrium is established between the capacitors.
- (1) $\frac{1}{6} CV_0^2$ (2) $\frac{1}{2} CV_0^2$
(3) $\frac{1}{3} CV_0^2$ (4) $\frac{1}{4} CV_0^2$
7. An empty conducting thick spherical shell possesses an inner radius R and an outer radius $2R$. Upon charging this hollow conductor with charge Q as indicated in the accompanying diagram, then the (V_A , V_B , V_C , and V_O are potentials at A , B , C and O respectively)

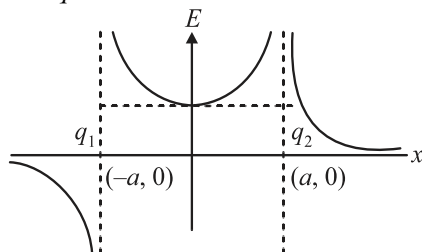


- (1) $V_C = V_O > V_A = V_B$
(2) $V_A > V_B > V_C > V_O$
(3) $V_A = V_B = V_C = V_O$
(4) $V_A = V_B < V_C < V_O$

8. A scientist was examining the electrostatic interaction between charged spheres. The setup comprised two concentric spheres, inner one solid and outer one hollow, each charged with equal but opposite charges. Initially, he measured the potential difference between the two spheres and recorded it as ϕ . Curious about the effect of altering the charges, he increased the charge on the hollow sphere to three times its original value. Now, he wonders how this change impacts the potential difference between the two spheres. What will the new potential difference be?

- (1) -3ϕ (2) ϕ
(3) 2ϕ (4) 9ϕ

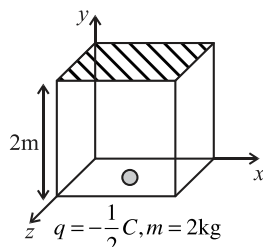
9. A graph between electric field and distance from the origin along x -axis is given for two charges q_1 and q_2 located at $x = -a$ and $x = +a$.



Assume that direction left to right is positive and right to left is negative. Which of the following statements is/are correct?

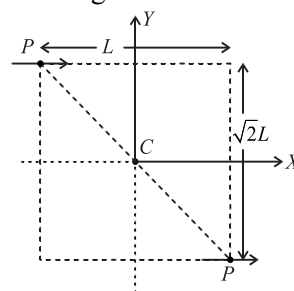
- (1) Both q_1 and q_2 are positive and $q_1 = q_2$
(2) q_1 is positive and q_2 is negative and $q_1 = q_2$
(3) q_1 is negative and q_2 is positive and $q_1 = q_2$
(4) Above curve is not possible

10. A closed container placed in a region devoid of gravity is depicted below. Initially, the system is stationary. Within the region, the electric potential varies according to $V = y^3 + 2$ (in joules per coulomb). A small object having charge q and mass m is released from rest at the base of the container and begins to ascend under the influence of the electric field. Eventually, it impacts the shaded inner surface of the container. If the ball's velocity immediately after the collision is 1.5 m/s and it remains in contact with the wall for 0.1 seconds, determine the external average force that must be applied to keep the container stationary during the impact, assuming the ball applies a constant force on the wall throughout the contact duration.



- (1) 70 N
(2) 72 N
(3) 74 N
(4) 76 N

11. Two identical point dipoles, each possessing a dipole moment of magnitude P , are positioned at two adjacent vertices of a rectangular configuration as illustrated. Considering their orientations as shown, the net direction of the electric field at the geometric center C of the rectangle is along

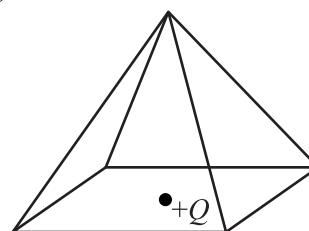


- (1) Negative Y -axis
(2) Positive Y -axis
(3) Positive X -axis
(4) Negative X -axis

12. A small charged body moves in a non-uniform electric field that varies along the x -direction according to $E = (1 - kx^2) \text{ N/C}$, where $k = 3 \text{ m}^{-2}$. The particle starts from rest at the coordinate origin ($x = 0$). Find the location on the x -axis (other than the initial position) where the particle comes to rest again.

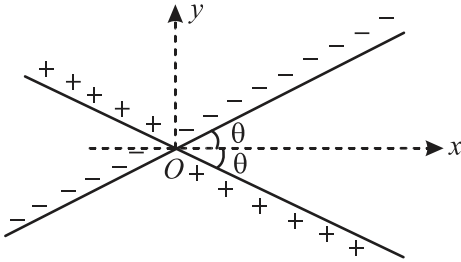
- (1) 3 m
(2) 2 m
(3) 2.5 m
(4) 1 m

13. A particle electric charge of magnitude $Q = 64\epsilon_0$ Coulombs sits at the midpoint of a quadrilateral floor of a pentahedral solid with four congruent triangular walls as shown. Determine the flux through any one triangular wall section. (Assume SI units)

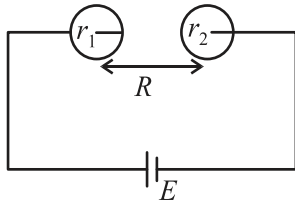


- (1) $4 \text{ N m}^2 \text{ C}^{-1}$
(2) $16 \text{ N m}^2 \text{ C}^{-1}$
(3) $8 \text{ N m}^2 \text{ C}^{-1}$
(4) $32 \text{ N m}^2 \text{ C}^{-1}$

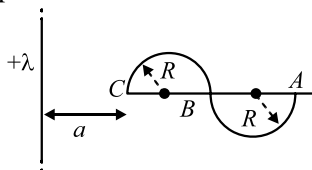
14. Consider two elongated uniform charge carriers of infinite extent having identical magnitude line charge distributions but with opposing polarity orientations as shown. The electric field at any coordinate position along the horizontal reference axis (x-axis) for positive values of x-coordinate follows the directional unit vector



- (1) $\cos \theta \hat{i} + \sin \theta \hat{j}$
 - (2) $\hat{i}$
 - (3) $\hat{j}$
 - (4) $-\sin \theta \hat{i} + \sin \theta \hat{j}$
15. A pair of spherical conductors with geometric measurements r_1 and r_2 positioned at separation distance R ($R \gg r_1, r_2$) are linked to a voltage source providing potential difference E . Excluding any influence from the connecting pathway, the magnitude of electrostatic force between these conductive spheres equals:

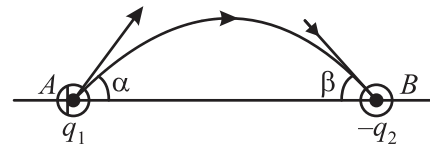


- (1) $\frac{16\pi^2 \epsilon_0^2 r_1^2 r_2^2}{R^2 (r_1 + r_2)^2}$
 - (2) $\frac{4\pi^2 \epsilon_0 E r_1 r_2}{R^2 (r_1 + r_2)^2}$
 - (3) $\frac{4\pi \epsilon_0 E^2 r_1^2 r_2^2}{R^2 (r_1 + r_2)^2}$
 - (4) $\frac{2\pi \epsilon_0 E r_1 r_2}{R^2 (r_1 + r_2)^2}$
16. In figure there is an infinitely long uniformly charged nonconducting wire having linear charge density $+\lambda$. Find the net work done in slowly taking a point charge q from A to C through semi-circular paths AB and BC .



- (1) $\frac{q\lambda}{\epsilon_0} \ln\left(\frac{a+2R}{a}\right)$
- (2) $\frac{q\lambda}{4\epsilon_0} \ln\left(\frac{a+4R}{a}\right)$
- (3) $\frac{q\lambda}{\pi\epsilon_0} \ln\left(\frac{a+2R}{a}\right)$
- (4) $\frac{q\lambda}{2\pi\epsilon_0} \ln\left(\frac{a+4R}{a}\right)$

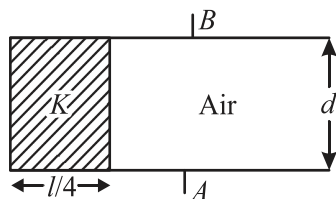
17. Consider two electrical sources $+q_1$ and $-q_2$ positioned at locations A and B correspondingly. An electrostatic field trajectory originates from q_1 with angular departure α relative to the AB reference line as shown. Determine the terminus angle β where this trajectory concludes at the $-q_2$



- (1) $\sin^{-1}\left[\sqrt{\frac{q_1}{q_2}} \sin \frac{\alpha}{2}\right]$
 - (2) $2\sin^{-1}\left[\sqrt{\frac{q_1}{q_2}} \sin \frac{\alpha}{2}\right]$
 - (3) $\sin^{-1}\left[\sqrt{\frac{q_2}{q_1}} \sin \frac{\alpha}{2}\right]$
 - (4) $2\sin^{-1}\left[\sqrt{\frac{q_2}{q_1}} \sin \frac{\alpha}{2}\right]$
18. A $6 \mu\text{F}$ capacitor links to a voltage source achieving potential difference 2 volt. A second $12 \mu\text{F}$ capacitor similarly reaches source voltage 4 volt. Now after the voltage sources disconnection, these two capacitors are connected across each other in opposing terminal configurations (positive to negative arrangement). Calculate the final energy of this combined system.
- (1) Zero
 - (2) $36 \mu\text{J}$
 - (3) $18 \mu\text{J}$
 - (4) $48 \mu\text{J}$
19. A capacitor is charged and then disconnect from the voltage source. Then the distance between the two plates is increased. Plates are still large in comparison to the distance between the plates.
- Statement I:** Attractive interaction force between plates diminishes.
- Statement II:** Voltage differential between plates increases.

- (1) Statement I is True, Statement II is True; Statement II is a correct explanation for Statement I.
- (2) Statement I is True, Statement II is True; Statement II is NOT a correct explanation for Statement I.
- (3) Statement I is True, Statement II is False.
- (4) Statement I is False, Statement II is True.

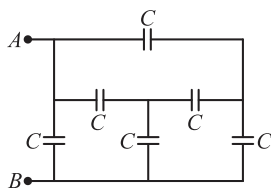
20. An energy storage configuration of two large square metallic parallel plates in between terminals A and B utilizing air ($K_{\text{air}} = 1$) as insulation barrier exhibits capacitance C . An insulating slab with dielectric constant K matching the inter-surface spacing partially occupies one-fourth of the geometric volume as depicted. Determine the new capacitance



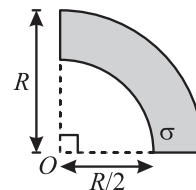
- (1) $(K+3)\frac{C}{4}$ (2) $(K+2)\frac{C}{4}$
 (3) $(K+1)\frac{C}{4}$ (4) $\frac{KC}{4}$

Integer Type Questions

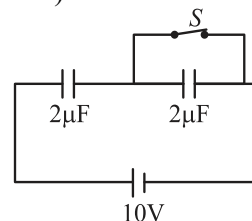
21. Equivalent capacitance (in μF) between the points A and B as shown in figure will be (Given $C = 12 \mu\text{F}$)



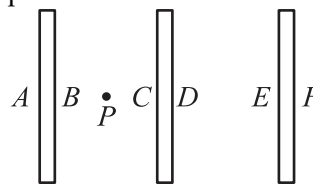
22. An illustrated geometric configuration displays a segment of a homogeneously electrified circular annular plate with inner and outer radii measurement $\frac{R}{2}, R$ respectively and exhibiting surface charge density σ . If the potential at the geometric midpoint O equals $\frac{1}{N} \times \frac{\sigma R}{2\epsilon_0}$, determine the numerical coefficient N .



23. Within the displayed electrical network configuration, the switch S begins in a conducting state. Calculate the thermal energy dissipation (measured in μJ) experienced by the complete system following the interruption (off state) of the switch. (Assume connecting wires are ideal with zero resistance)



24. Three large parallel charged conducting plates are placed a small distance d apart from each other. The surface charge density on the face B of a conductor is $+5.0\epsilon_0 \text{ C/m}^2$ and that on face E is $+6.0\epsilon_0 \text{ C/m}^2$. Charges on other faces are not known. Find the magnitude of electric field intensity at P (shown in the figure) between the plates in N/C .



25. There is a fixed circular ring of radius 20 cm carrying charge $Q = +10 \mu\text{C}$ uniformly distributed on its periphery and a small mass of 20 grams having charge $q = -10 \mu\text{C}$ can move along the axis of the fixed ring. If the mass is disturbed slightly from the centre of the ring then it executes SHM. Find the angular frequency (in rad/s) of mass about centre of the circular charge.

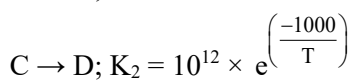
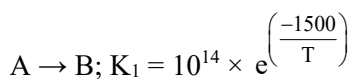
SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. K_f for water is $1.86 \text{ K kg mol}^{-1}$. If automobile radiator holds 2.0 kg of water, how many grams of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) must be added to get the freezing point of the solution lowered to -2.6°C ?
 (1) 27g (2) 173 g
 (3) 93 g (4) 69 g
27. Two liquid A and B have vapour pressure in the ratio $P_A^0 : P_B^0 = 1 : 3$ at a certain temperature. The mole fractions of A to B in the vapour phase is

4 : 3, then the mole fraction of B in the solution at the same temperature is

- (1) $\frac{1}{5}$
 (2) $\frac{2}{3}$
 (3) $\frac{4}{5}$
 (4) $\frac{1}{4}$
28. For the reaction, following data is given,



The temperature at which $K_1 = K_2$ is :

- (1) 118.55 K (2) 108.55 K
(3) 468.4 K (4) 434.22 K

29. Consider following cases

- (I) 2M CH_3COOH solution in benzene at 27°C where there is dimer formation to the extent of 50%.
(II) 0.5 M NaCl aq. solution at 27°C , which ionises 100%.

Which of the following is true statement?

- (1) Both are isotonic
(2) I is hypertonic w.r.t. II
(3) II is hypertonic w.r.t. I
(4) I has less elevation in boiling point than II (considering same k_b values and $M \approx m$)

30. In first order reaction initial concentration of reactant is 15M. Find the time in which concentration is reduced up to 5M and half life of reaction is 5 Min.

- (1) 8 minute (2) 10 minute
(3) 20 minute (4) 24.6 minute

31. Consider the following data for the reaction between A and B

S.No.	[A] mol.L^{-1}	[B] mol.L^{-1}	Rate $\text{mol.L}^{-1} \text{sec}^{-1}$
I	1×10^{-2}	2×10^{-2}	2×10^{-4}
II	2×10^{-2}	2×10^{-2}	4×10^{-4}
III	2×10^{-2}	4×10^{-2}	8×10^{-4}

Which of the following are correct statements

- (a) unit of rate constant of the reaction is $(\text{mol})^{1-n} (\text{L})^{n-1} \text{sec}^{-1}$, where n is the order of the reaction
(b) Rate law of the reaction is $k[A][B]^2$
(c) Rate of reaction increases eight times on doubling the concentration of both the reactant.
(1) a, b and c (2) a and b
(3) b and c (4) a alone

32. At 25°C , the total pressure of a solution obtained by mixing 2 moles of A and 3 moles of B is 168 torr. What is the vapour pressure (in torr) of pure A at the same temperature?

(Vapour pressure of pure B, at 25°C is 140 torr)

- (1) 180 (2) 160
(3) 210 (4) 200

33. Find the ratio of time $t_{1/4}$ to $t_{7/8}$ for zero order reaction.

- (1) 4 : 15 (2) 2 : 15
(3) 1 : 7 (4) 2 : 7

34. The molar conductivity of acetic acid at a given concentration is $\lambda_{\text{CH}_3\text{COOH}} = 27 \text{ ohm}^{-1} \text{cm}^2 \text{mol}^{-1}$

while its limiting molar conductivity is $\lambda_{\text{CH}_3\text{COOH}}^0 = 540 \text{ ohm}^{-1} \text{cm}^2 \text{mol}^{-1}$. For a 1 M

Solution of this acid, calculate the pH

- (1) 1.7 (2) 1.3
(3) 0 (4) 4

35. 20 gm of CH_3COOH is dissolved in 1000 gm of each of the following solvents. In which solvent, the mole fraction of solute is maximum?

(Assume No dissociation or association of CH_3COOH)

- (1) Water (2) Ethanol
(3) Benzene (4) Same in all

36. For a chemical reaction $3\text{O}_2 \rightarrow 2\text{O}_3$ the rate of disappearance of O_2 is $0.06 \text{ mole lit}^{-1} \text{sec}^{-1}$. Determine the rate of formation of O_3 :

- (1) $0.04 \text{ mol lit}^{-1} \text{sec}^{-1}$
(2) $0.08 \text{ mol lit}^{-1} \text{sec}^{-1}$
(3) $0.10 \text{ mol lit}^{-1} \text{sec}^{-1}$
(4) $0.06 \text{ mol lit}^{-1} \text{sec}^{-1}$

37. The reduction potential of hydrogen half cell will be negative if:-

- (1) $P_{\text{H}_2} = 1 \text{ atm}; [\text{H}^+] = 2.0 \text{ M}$
(2) $P_{\text{H}_2} = 1 \text{ atm}; [\text{H}^+] = 1 \text{ M}$
(3) $P_{\text{H}_2} = 4 \text{ atm}; [\text{H}^+] = 1 \text{ M}$
(4) $P_{\text{H}_2} = 2 \text{ atm}; [\text{H}^+] = 2 \text{ M}$

38. A and B form ideal solution, at 90°C , P_A^0 is half of P_B^0 . A solution containing 0.2 mole of A and 0.8 mole of B has a normal boiling point of 90°C . find the value of P_B^0 :

- (1) $\frac{10}{9} \text{ atm}$
(2) $\frac{5}{9} \text{ atm}$
(3) $\frac{15}{9} \text{ atm}$
(4) $\frac{20}{9} \text{ atm}$

39. For a chemical reaction, $A + 2B \rightarrow C + D$, the rate of reaction increases three times, when concentration of A only is increased to nine times. While when concentration of B only is increased 2 times, then rate of reaction also becomes 2 times. The order of this reaction is
- (1) 3 (2) $3/2$
(3) $1/2$ (4) None of these

40. An aqueous solution of a salt MX_3 at certain temperature has a van't Hoff factor of 3.4. The degree of dissociation for this solution of the salt is
- (1) 0.50 (2) 0.80
(3) 0.67 (4) 0.33

41. If the standard reduction potential, E° for four divalent metals X, Y, Z, W are -1.46 V, -0.36 V, 0.15 V and -1.24 V respectively, then
- (1) X will displace Z^{+2} from aqueous solution
(2) Y will displace W^{+2} from aqueous solution
(3) W will displace X^{+2} from aqueous solution
(4) Z will displace Y^{+2} from aqueous solution

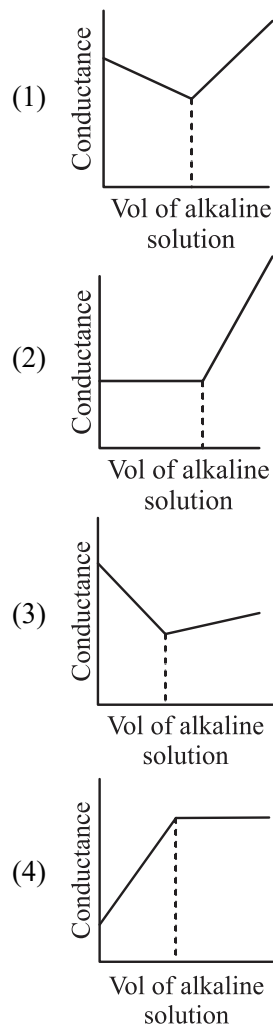
42. Match the column

Column-I (Reduction process)		Column-II (Charge required)	
A	1 mole of MnO_4^- to Mn^{2+}	P	193000 C
B	1 mole of $Cr_2O_7^{2-}$ to Cr^{3+}	Q	289500 C
C	1 mole of Sn^{4+} to Sn^{2+}	R	482500 C
D	1 mole of Al^{3+} to Al	S	579000 C

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (1) P | R | Q | S |
| (2) R | S | P | Q |
| (3) S | Q | P | R |
| (4) R | S | Q | P |

43. The reduction potential of a half-cell consisting of a Pt electrode immersed in 1 M Fe^{2+} and 0.1 M Fe^{3+} solution at $25^\circ C$ is
- ($E^\circ_{Fe^{3+}/Fe^{2+}} = 0.770$ V)
- (1) 0.83 V
(2) 0.88 V
(3) 0.71 V
(4) 0.65 V

44. Identify the plot which is for a conductometric titration of strong acid Vs weak base



45. During the electrolysis of an acetate solution ethane is formed according to the Kolbe reaction $2CH_3COO^- \rightarrow C_2H_6 + 2CO_2 + 2e^-$. What volume of ethane is produced at $27^\circ C$ and 720 mm Hg, if a current of 0.4 ampere were passed through the solution for 8 hours and the electrode reaction is 80% efficient:-
- (1) 2.7 L (2) 1.24 L
(3) 1.03 L (4) 0.65 L

Integer Type Questions

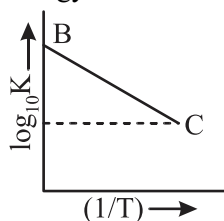
46. At certain temperature (T) if conductivity of pure water is $4.3 \times 10^{-7} \text{ Scm}^{-1}$, then calculate pH of water at given temp.

Given: $\lambda^\infty_{H^+} = 290 \text{ Scm}^2 \text{eq}^{-1}$

$\lambda^\infty_{OH^-} = 140 \text{ Scm}^2 \text{eq}^{-1}$

47. A solution is prepared by mixing of isopropyl alcohol and propyl alcohol. It has a vapour pressure of 250 and 300 mm of Hg when the mole percentage of isopropyl alcohol is 25% and 75% respectively. The vapour pressure of Propyl alcohol in mm of Hg is _____

48. Consider the following graph of $\log_{10}K$ vs $\frac{1}{T}$ where K is rate constant and T is temperature. The straight line BC has slope, $\tan\theta = -\frac{40}{2.303}$ and an intercept of 8 on Y-axis. Thus E_a , the energy of activation is (in calorie)



49. Two substances A ($t_{1/2} = 4$ mins) and B ($t_{1/2} = 6$ mins) follow first order kinetics are taken in such a way that initially $[A] = 8[B]$. Calculate the time (in min) after which the concentration of both the substance will be equal.
50. An ideal solution was found to have a vapour pressure of 180 torr when the mole fraction of a non-volatile solute was 0.1. What would be the vapour pressure of the pure solvent (in torr) at the same temperature?

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. If $f(r) = \begin{bmatrix} 3r & r+1 \\ 2r & r+2 \\ 3r & r+7 \end{bmatrix} \begin{bmatrix} 1 & r & r^2 \\ 1 & r^2 & r^4 \end{bmatrix}$ then $\sum_{r=1}^{2025} |f(r)|$ equals (where $|A|$ denotes determinant of A)
- (1) 2025 (2) 0
(3) $(2025)^3$ (4) $(2025)^2$

52. The number of solutions of the equation $|2x - 1| = 3[x] + 2\{x\}$ (where $[.]$ denotes the G.I.F. and $\{t\}$ represents the fractional part of (t)) is
- (1) 2 (2) 1
(3) 3 (4) 5

53. The fundamental period of $f(x) = 4\cos^4\left(\frac{\sqrt{2}x - \pi}{4\pi^2}\right) - 2\cos\left(\frac{\sqrt{2}x - \pi}{2\pi^2}\right)$ is
- (1) $2\sqrt{2}\pi^3$ (2) $\sqrt{2}\pi^3$
(3) $2\pi^3$ (4) Non-periodic

54. The $f(x) = \frac{x^3}{3} + (m-1)x^2 + (m+5)x + 11$ is invertible only for 'k' integral values of 'm' then 'k' is
- (1) 4 (2) 6
(3) 7 (4) 8

55. Let $\Delta_1 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} bc-a^2 & ac-b^2 & ab-c^2 \\ ac-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ac-b^2 \end{vmatrix}$.

If Δ_1, Δ_2 is divisible by $(a+b+c)^\lambda$ then maximum value of λ is

- (1) 1 (2) 2
(3) 3 (4) 4

56. Let the function $f: D_f \rightarrow R - \left[-1, -\frac{1}{3}\right]$

$$f(x) = \frac{x-1}{p-x^2+1} \text{ then}$$

- (1) p can not be a negative integer
(2) p must be positive
(3) p cannot take any positive integral values
(4) p can not take any integral values

57. Consider,

$$f(n) = \begin{vmatrix} 2 & 1 & 0 \\ \frac{1}{(n+3)^2} & \frac{1}{n+1} & \frac{1}{(n+3)^2} - \frac{1}{(n+1)} \\ \frac{1}{(n+2)^2} & \frac{1}{n+2} & \frac{-(n+1)}{(n+2)^2} \end{vmatrix}$$

$$\text{where } n \in N \text{ then } \sum_{n=1}^7 f(n) - \sum_{n=1}^{\infty} f(n) =$$

- (1) $\frac{1}{18}$ (2) $-\frac{1}{900}$
(3) $\frac{1}{450}$ (4) $\frac{49}{450}$

58. The function $f: [0, 2] \rightarrow [1, 4]$, defined by $f(x) = x^3 - 5x^2 + 7x + 1$, is
- (1) one-one onto
(2) onto but not one-one
(3) one-one but not onto
(4) neither one-one nor onto

59. For each positive integer n , let $f(n+1) = n(-1)^{n+1} - 2f(n)$ and $f(1) = f(2010)$.

$$\text{Then } \sum_{k=1}^{2009} f(k) \text{ is equal to}$$

- (1) 335 (2) 336
(3) 331 (4) 333

60. Let $A = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 5 & 2 \\ 2 & 1 \end{bmatrix}$ and $C = (A(ABA^T)^{25}(AB^T A^T)^{10} A^T)^T$, If trace of matrix C is λ and $|C| = \mu$ then $\frac{\lambda}{\mu}$ is

- (1) 2 (2) 4
(3) 6 (4) 8

61. The sum of positive integral values of x satisfying the equation $\text{sgn}\left(\left[\frac{197}{1+x^4}\right]\right) = [1 + \{2025x\}]$ is

$$\left(\left[\frac{197}{1+x^4}\right]\right) = [1 + \{2025x\}]$$

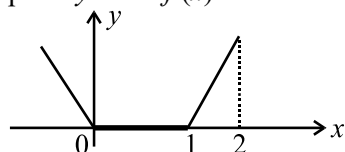
(Note: $\text{sgn}(y)$, $[y]$, $\{y\}$ denote signum function, greatest integer function and fractional part function respectively.)

- (1) 6 (2) 7
(3) 0 (4) 16

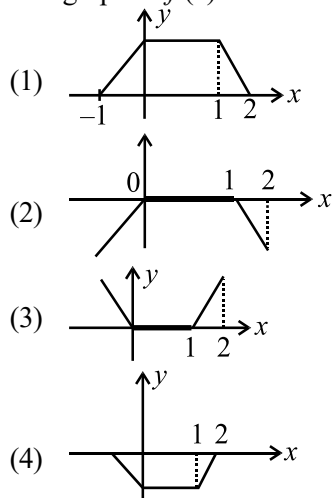
62. Let R_1 and R_2 be two relations defined on R by $aR_1b \Leftrightarrow ab \geq 0$ and $aR_2b \Leftrightarrow a \geq b$ then $S_1 : R_1$ is not equivalence because it is not symmetric
 $S_2 : R_2$ is not equivalence because it is not transitive

- (1) S_1 is true and S_2 is false
(2) S_1 is false and S_2 is true
(3) Both S_1 and S_2 are true
(4) Both S_1 and S_2 are false

63. Let graph of $y = 1 - f(x)$ is shown below



The graph of $f(x)$ will be:



64. Let A be a 2×2 matrix defined as $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ then A^{-1} is
- (1) $I + \frac{1}{5}A$ (2) $I - \frac{1}{5}A$

- (3) $\frac{1}{5}A$ (4) $-\frac{1}{5}A$

65. If p, q, r are distinct non zero real numbers and

$$f(x) = \begin{vmatrix} p^2 + x^2 & pq & pr \\ pq & q^2 + x^2 & qr \\ pr & qr & r^2 + x^2 \end{vmatrix}, \text{ then the}$$

number of distinct (both real and imaginary included) roots of $f(x) = 0$ are

- (1) 4
(2) 5
(3) 2
(4) 3

66. Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ are two

matrices. If product of A and B commute, then which of the following is true?

- (1) $2b = 3c$
(2) $a + c = d$
(3) $2a + 2d = 3c$
(4) $a + b = c + d$

67. If $f(x) = x^3 + x + 1$, then integral part of the real number $f^{-1}(4)$ must be

- (1) -1
(2) 1
(3) 2
(4) 69

68. If $a \neq p, b \neq q, c \neq r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$ then the

$$\text{value of } \frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c}$$

- (1) 0 (2) 2
(3) -1 (4) 3

69. The value of λ can not take for which the following system of equations have unique solution:

$$\begin{aligned} x + y + z &= 3 \\ 2x - y + z &= 2 \\ x + \lambda y - z &= 3 \end{aligned}$$

- (1) 1 (2) 2
(3) -5 (4) 3

70. If $f(x) + 2f(1-x) = x^2 + 2; \forall x \in R$, then $f(x)$ is

- (1) Invertible function
(2) Periodic function
(3) Even function
(4) Neither even nor odd function

Integer Type Questions

71. Let the number of 4×4 matrices $A = [a_{ij}]_{4 \times 4}$ whose entries are each '2025' or '-2025' such that the sum of the entries in each row and in each column vanishes and $a_{11} \geq a_{ij} \forall i, j \in \{1, 2, 3, 4\}$ are " $15k$ ", then k equals to

72. If $f(x) = \begin{vmatrix} \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \sin(\beta-\gamma) & \sin(\gamma-\beta) & \sin(\alpha-\beta) \end{vmatrix}$
and $f(0) = \frac{1}{4}$, then $\left[\sum_{r=1}^{15} f(x_r) \right]$ is (where $[.]$ is G.I.F.)

73. If $n(P(A)) = 16$, then total number of symmetric relations which one not reflexive relations that can be defined from $A \rightarrow A$ is $\alpha(2^\gamma)$, then ' $\alpha - \gamma$ ' is

74. If

$g(x) = \left(4\sin^4 x - 6\sin^2 x + 2 - 2\cos^2 x - \frac{1}{2}\cos 4x - x^9 \right)^{1/9}$,
then the sum of digits of $g(g(2025))$ is

75. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 3 \\ 1 & 0 \end{bmatrix}$ and

$X = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$. If $P = AXB^T$ and

$Q = B^T X^T A$, then number of positive integral divisors of $\det(PQ)$ is